# Changing velocity

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Calculate acceleration and distinguish speeding up, slowing down, and turning.

**SUCCESS CRITERIA - I can...**

1. Calculate acceleration from velocity change and time.
2. Compare acceleration direction with velocity direction.
3. Explain why turning at steady speed is acceleration.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: read motion in two ways

(a) A remote-control car travels west at 4 m/s for 6 s. How far west does it move?
(b) A probe's velocity has components of 8 m/s east and 3 m/s south on a tracking map. Sketch the components and the probe's overall direction.

**Marzano check - circle:** ① ② ③

## 02 Worked example: braking on a bike

**Reference:** $a_{avg} = (v_f - v_i)/\Delta t$.
$v_i$ and $v_f$ are initial and final velocity. $\Delta t$ is elapsed time. With a steady change in velocity, this gives the constant acceleration.

**Worked reasoning:** A cyclist brakes while riding east. Her speed drops steadily from 8 m/s to 2 m/s in 3 s. Take east as positive.
The velocity change is 2 - 8 = -6 m/s.
Acceleration = -6 ÷ 3 = -2 m/s².
She is still moving east, but her acceleration points west.

**Your turn:** The model shows the cyclist's velocity before and after braking. Add a separate acceleration arrow below it. Explain why she is slowing down even though both velocity arrows point east.

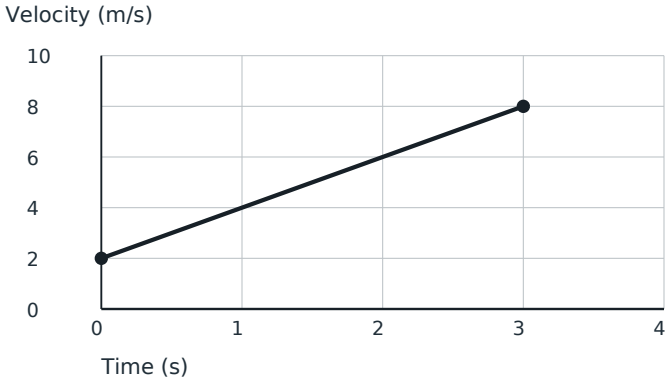



Acceleration: add a separate arrow.

**Marzano check - circle:** ① ② ③

## 03 A remote-control car speeds up

The graph shows a remote-control car during a straight-track test. East is positive.
(a) Read its velocity at 0 s and 3 s.
(b) Calculate the change in velocity, then the acceleration.
(c) What feature of the graph shows that its speed is increasing?




**Marzano check - circle:** ① ② ③

## 04 Slowing before a crosswalk

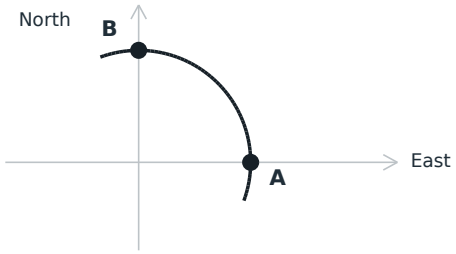
A cyclist riding east slows steadily from 10 m/s to 4 m/s as she approaches a crosswalk. This takes 5 s.
(a) How much speed does she lose each second?
(b) Find her acceleration, including its direction. Is she still moving east?

**Support:** Find the total speed lost first. Divide by the elapsed time. Compare the direction of acceleration with the direction of motion.

**Marzano check - circle:** ① ② ③

## 05 Running around a bend

A runner goes from A toward B around a bend at a steady speed. A friend says, "If the speed stays the same, there is no acceleration."
Draw a velocity arrow at A and at B on the path below. Do you agree with the friend? Use the arrows to explain.

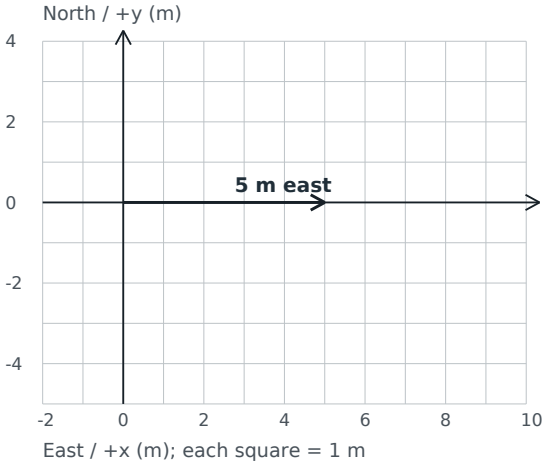



Path sketch: runner moves from A toward B.

**Marzano check - circle:** 1 2 3

## 06 Review: a robot's route

During a navigation test, a robot drives 5 m east, 3 m north, 2 m west, then 7 m south.
(a) The first move is shown. Complete the route on the grid.
(b) Draw an arrow from the start to the endpoint. How far east or west, and north or south, does it finish from its start?




**Marzano check - circle:** 1 2 3

## 07 Exit check: a docking simulation

In a docking simulation, a spacecraft's velocity changes steadily over 2 s. The arrows show its velocity at the start and end.
(a) Read the speeds. Is the spacecraft speeding up or slowing down?
(b) Calculate its acceleration. Add an acceleration arrow and explain its direction.

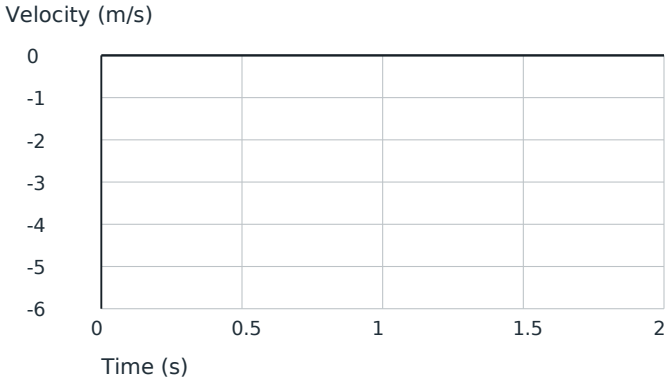



Acceleration: add a separate arrow.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: negative can mean faster

In a straight-track test, a remote-control car speeds up steadily from 1 m/s west to 5 m/s west in 2 s. East is positive.
Complete the velocity-time graph below. Calculate acceleration and explain how a negative acceleration can make this car speed up.




**Marzano check - circle:** 1 2 3